AC2204 Marking Scheme

Introduction to Management Accounting · Autumn 2020 · Paper 1

Examination Session and Year	Autumn 2020
Module Code	AC2204
Module Title	Introduction to Management Accounting
Paper Number	Paper 1
External Examiner	Professor Ivo De Loo
Head of the Department	Professor Mark Mulcahy
Internal Examiners	Dr Margaret Healy
Duration of Paper	1.5 hours
Special Requirements	Show your workings; marks will be awarded for workings. The use of a non-programmable calculator is permitted. Separate answer books are not required.

Instructions to Candidates (as per exam paper): Answer three questions in total. Attempt Question One from Section A (compulsory) and TWO questions from Section B.

Own-figure rule: where a candidate makes an error at an early stage of a multi-step numerical question, full method marks are awarded for correctly applying the required technique to their own (incorrect) figure in all subsequent steps. Marks are never cascaded away for a single early error.

Marking philosophy: this is a closed-book, time-pressured examination. Academic citations are never required. Discursive answers are marked on depth of explanation and application to the specific scenario, not generic textbook recall.

Question Overview

Question	Scenario	Topics	Marks
Q1 (compulsory)	Orla O'Gorman (eco T-shirts)	CVP analysis, breakeven, CVP graph, degree of operating leverage, cost behaviour theory (fixed/variable/mixed)	40
Q2 (Section B option)	Con's Garage	Financial vs management accounting, high-low method, ABC cost hierarchy	30
Q3 (Section B option)	Asia Ltd	Variable (marginal) vs absorption costing, full profit and loss accounts, separately variable/fixed selling costs	30
Q4 (Section B option)	Statement evaluation	Prime vs conversion costs, breakeven/MOS/operating leverage verification, differential costs, fixed cost per unit, relevant range	30

QUESTION 1

Orla O'Gorman – Eco T-Shirts

Total: 40 marks

Orla O'Gorman has set up a business producing eco-friendly T-shirts for sale at festivals, concerts and sporting events. Candidates are required to (A) calculate the contribution margin ratio and breakeven point, with a CVP graph, (B) calculate the degree of operating leverage and use it to advise Orla on a proposed 30% sales increase, and (C) discuss a statement about cost behaviour, with diagrams.

Part	Requirement	Marks
(A)	Contribution margin ratio, breakeven point (units and revenue), CVP graph	20
(B)	Degree of operating leverage and its use in advising on a 30% sales increase	8
(C)	Discuss "all costs change directly in proportion to units produced", with diagrams	12

Question 1(A) **20 marks**

Orla O'Gorman, eco-friendly T-shirts: Sales price per unit €24; Variable expenses per unit €13; Fixed costs: Rental €12,000, Depreciation €3,000, Advertising €2,000, Administration €5,000. (A) Calculate the contribution margin ratio and the breakeven point in both units and sales revenue. Prepare a CVP graph, indicating the breakeven point.

Full Model Answer

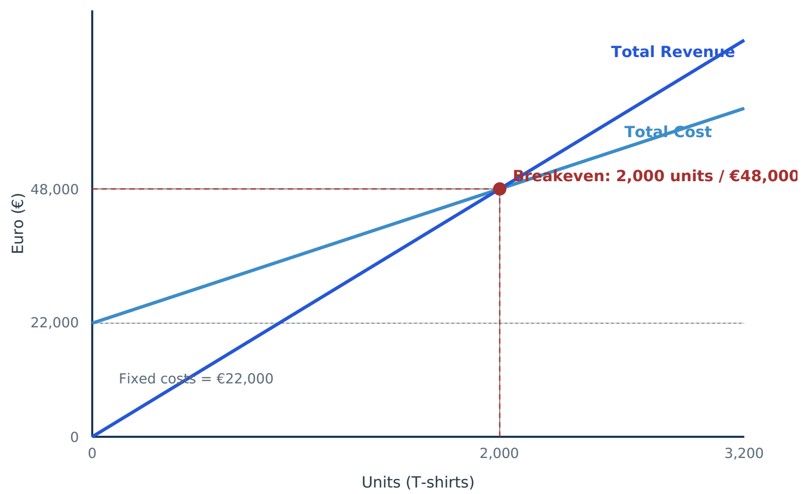
Working – total fixed costs and contribution margin

Item	€
Rental of business start-up unit	12,000
Depreciation of printing equipment	3,000
Advertising costs	2,000
Administration costs	5,000
Total fixed costs	**22,000**
Contribution margin per unit (€24 − €13)	11.00
Contribution margin ratio (€11 ÷ €24)	**45.83%**

Breakeven point

Item	Value
Breakeven units (€22,000 ÷ €11)	**2,000 T-shirts**
Breakeven revenue (2,000 × €24, or €22,000 ÷ 45.83%)	**€48,000**

CVP graph:



Graph description: The total revenue line starts at the origin (€0 at 0 units) and rises with a slope of €24 per unit. The total cost line starts at the €22,000 fixed cost intercept (the cost incurred even at zero sales) and rises with a slope of €13 per unit. The breakeven point is where the two lines cross, at 2,000 units / €48,000 — below this point, total cost exceeds total revenue (a loss); above it, total revenue exceeds total cost (a profit).

Mark Allocation

Marks	Criteria
18–20	Contribution margin ratio and breakeven point (units and revenue) correctly calculated; CVP graph correctly shows both lines with the correct intercepts and slopes, and the breakeven point clearly marked and labelled.
13–17	Calculations correct; graph substantially correct with a minor labelling or scaling issue.
8–12	Calculations correct; graph significantly incomplete or missing key features (e.g. no fixed cost intercept, breakeven point not marked).
0–7	Limited understanding of CVP calculations or graphing; minimal correct workings.

Question 1(B) **8 marks**

During its first year in operation, the business sold 2,500 T-shirts. Orla is confident that with a more intense sales effort and a more creative advertising campaign, she can increase sales volume by 30% next year. (i) Determine the degree of operating leverage for the business. (ii) Explain to Orla what this calculation means and use your answer to advise her on the expected amount of total profit resulting from a 30% increase in sales.

Full Model Answer – (i) Degree of operating leverage

Working

Item	€
Total contribution margin (2,500 units × €11)	27,500
Less: Total fixed costs	(22,000)
Operating profit	**5,500**
Degree of operating leverage (€27,500 ÷ €5,500)	**5.0**

Full Model Answer – (ii) Explanation and advice

> **What a degree of operating leverage of 5.0 means**
> The degree of operating leverage measures how sensitive operating profit is to a change in sales. A DOL of 5.0 means that for every 1% change in sales revenue (or volume), operating profit is expected to change by approximately 5%. This sensitivity arises because Orla's fixed costs (€22,000) remain constant regardless of volume, so once they are covered, each additional T-shirt sold contributes disproportionately to profit growth — but equally, a sales *decline* would hit profit hard in percentage terms.

Applying the DOL to a 30% sales increase

Item	Value
Expected % change in operating profit (30% × DOL of 5.0)	150%
Current operating profit	€5,500
Expected operating profit next year (€5,500 × (1 + 150%))	**€13,750**

Cross-check: This can be verified directly: a 30% increase takes sales from 2,500 to 3,250 units. Total contribution margin = 3,250 × €11 = €35,750; less fixed costs of €22,000 gives an operating profit of €13,750, confirming the DOL-based estimate exactly (this holds precisely, not just approximately, provided the cost structure — selling price, variable cost per unit, and fixed costs — remains unchanged).

Mark Allocation

Marks	Criteria
7–8	DOL correctly calculated as 5.0; clear explanation of what it measures; correct application to estimate the expected profit of €13,750 from a 30% sales increase, with a supporting cross-check.
5–6	DOL correctly calculated; explanation reasonable; profit estimate correct or with a minor error.
2–4	DOL calculated with an error, or explanation is generic without clear application to the 30% increase.
0–1	No meaningful calculation or explanation provided.

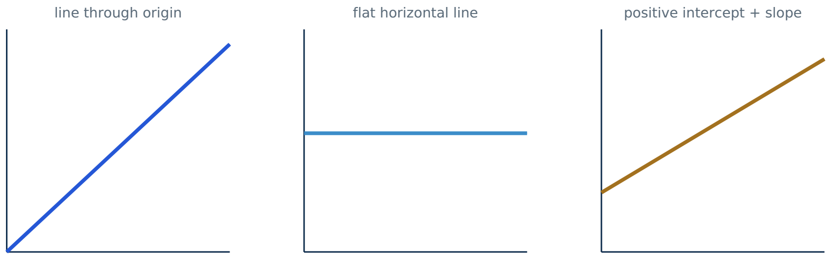
Question 1(C) **12 marks**

"All costs change directly in proportion to increases in the number of units produced". Discuss this statement, using diagrams to illustrate your answer.

Full Model Answer

DISAGREE

This statement is only true of variable costs. Not all costs behave this way — fixed costs do not change at all with volume, and mixed costs change, but not in direct proportion.



Variable cost Fixed cost Mixed cost

Variable costs – the statement is true here

A variable cost does change directly in proportion to volume: total variable cost is zero at zero activity and rises in a straight line through the origin, with a constant variable cost per unit (e.g. Orla's €13 variable cost per T-shirt). This is the only one of the three cost types the statement correctly describes.

Fixed costs – the statement is false here

A fixed cost does not change at all as volume changes within the relevant range; total fixed cost is a flat, horizontal line (e.g. Orla's €22,000 of rent, depreciation, advertising and administration costs are incurred in full even if she sells zero T-shirts). Far from changing in direct proportion to units produced, fixed costs are completely unresponsive to volume in the short term.

Mixed (semi-variable) costs – the statement is also false here

A mixed cost contains both a fixed and a variable element: it does not start at zero (there is a positive cost even at zero activity, the fixed portion) and it does not rise in strict proportion to volume, since only the variable portion of the cost increases with each additional unit while the fixed portion remains constant. Graphically, a mixed cost is a straight line with a positive vertical intercept and a positive (but less steep, proportionally) slope than a pure variable cost.

Conclusion: The statement conflates all costs with variable costs specifically. A complete costing system needs to recognise and separately account for all three (or four, including step costs) patterns of cost behaviour — treating every cost as if it varied directly with volume would badly distort product costing, CVP analysis and short-term decision-making.

Mark Allocation

Marks	Criteria
10–12	Correct disagreement; all three cost behaviours (variable, fixed, mixed) correctly and clearly distinguished with accurate diagrams for each, explaining precisely why the statement only holds for variable costs.
7–9	Correct disagreement with accurate discussion of at least two cost types and diagrams; the third less developed.
4–6	Some relevant discussion of cost behaviour but incomplete, generic, or diagrams missing/inaccurate.
0–3	Agrees with the statement without meaningful justification, or no meaningful discussion provided.

QUESTION 1 TOTAL — 40 MARKS

QUESTION 2

Con's Garage

Total: 30 marks

Con's Garage services cars, with activity level and cost data available for 2018. Candidates are required to (A) compare financial and management accounting for a small business context, (B) apply the high-low method to estimate 2020 costs, and (C) describe the ABC cost hierarchy with garage-specific examples at each level.

Part	Requirement	Marks
(A)	Differences and similarities between financial and management accounting for Con's Garage	10
(B)	High-low cost function and expected total cost for 400 cars	10
(C)	ABC cost hierarchy with garage-specific examples at each level	10

Question 2(A)

10 marks

Identify and describe the main differences and similarities between financial accounting and management accounting for the management of Con's Garage.

Full Model Answer

Similarities

Both financial and management accounting draw on the same underlying transactional data (invoices for parts, wages paid to mechanics, receipts from customers) recorded within Con's Garage's accounting system. Both ultimately aim to provide financial information that supports the successful running of the business, and both express information in monetary terms to allow comparison and aggregation.

Differences – users and purpose

Financial accounting produces the annual accounts required for tax authorities, the bank (if Con has a loan), and potentially other external parties, giving a reliable overall picture of the garage's performance and position. Management accounting produces information for Con himself, to help him make day-to-day and strategic decisions — for example, what to charge for a car service, whether to take on an apprentice, or how many cars need to be serviced each month to break even (as calculated in Question 2(B)).

Differences – regulation, format and detail

Financial accounting is governed by tax law and accounting requirements dictating what must be reported and how. Management accounting is entirely unregulated: Con can produce information in whatever format and at whatever level of detail is useful — for example, cost per car serviced, cost by type of repair, or a weekly cash flow forecast — none of which financial accounting would require or provide.

Differences – time orientation

Financial accounting is historical, reporting on a period that has already ended. Management accounting is substantially forward-looking, supporting planning and budgeting for future periods (such as estimating costs for January 2020, as required in part (B)).

Mark Allocation

Marks	Criteria
8–10	At least one relevant similarity and three distinct, well-explained differences (users, regulation/format, time orientation) given, applied specifically to Con's Garage.
5–7	Similarities and differences identified with reasonable explanation; less specifically applied to Con's Garage.
2–4	Some relevant points made but generic or incomplete.
0–1	Minimal or no relevant discussion.

Question 2(B)

10 marks

Activity levels and associated costs at Con's Garage, 2018: HIGH: August, 800 cars serviced, total cost €24,040. LOW: November, 250 cars serviced, total cost €13,700. Determine the expected total costs if it is planned to service 400 cars in January 2020, based on the cost formula for 2018. State the cost function for 2018.

Full Model Answer

Working – high-low method

Item	Value
Variable cost per car ((€24,040 − €13,700) ÷ (800 − 250))	(€10,340 ÷ 550) = €18.80 per car
Fixed cost (at high point: €24,040 − (800 × €18.80))	€9,000
Fixed cost (at low point: €13,700 − (250 × €18.80)) – check	€9,000

Cost function for 2018

Item	Value
Cost function	**Total cost = €9,000 + (€18.80 × cars serviced)**

Estimated total costs for January 2020 (400 cars)

Item	Value
€9,000 + (€18.80 × 400)	€9,000 + €7,520
Estimated total costs	**€16,520**

Caveat: This estimate assumes 2018's cost structure (fixed cost of €9,000 and variable cost of €18.80 per car) remains valid into January 2020, which may not hold given the two-year gap — inflation in parts and labour costs, for example, would likely mean actual January 2020 costs are somewhat higher than this estimate.

Mark Allocation

Marks	Criteria
8–10	Variable cost per car and fixed cost correctly calculated and cross-checked at both points; cost function correctly stated; estimate for 400 cars correctly calculated as €16,520.
5–7	Substantially correct with a minor computational error, or the cross-check omitted.
2–4	High-low method attempted with the correct general approach but a significant computational error.
0–1	Little or no understanding of the high-low method.

Question 2(C) **10 marks**

Describe the ABC (activity-based costing) cost hierarchy, listing examples of typical costs relevant to Con's Garage for each level.

Full Model Answer

The ABC cost hierarchy, applied to Con's Garage

Level	Description	Example at Con's Garage
Unit-level	Costs incurred once for every single unit of output (here, every car serviced); vary directly and proportionally with the number of cars serviced.	Oil, filters, and consumable parts used on each individual car service.
Batch-level	Costs incurred once for a batch or group of units, regardless of how many units are in the batch; do not vary with the number of individual units within a batch.	Cost of placing a single bulk order for spare parts covering several cars' services; time spent setting up diagnostic equipment before working through a batch of similar jobs.
Product/service-sustaining level	Costs incurred to support a particular product or service line as a whole, regardless of how many units of that line are produced/serviced.	Maintaining specialised equipment dedicated to one type of service (e.g. electric vehicle diagnostic equipment); staff training/certification specific to a particular repair specialism.
Facility-sustaining level	Costs incurred to support the organisation as a whole; cannot be traced to any specific unit, batch, or service line at all.	Rent of the garage premises, business insurance, utility bills, and Con's own management/administration time.

Why the hierarchy matters: Traditional volume-based costing (e.g. a single rate per labour hour) implicitly treats all overhead as if it were unit-level, distorting costs for services that actually consume batch-level, product-sustaining, or facility-sustaining resources disproportionately. The ABC hierarchy makes explicit that different types of cost are driven by different things, allowing more accurate tracing of cost to each service.

Mark Allocation

Marks	Criteria
8–10	All four levels of the ABC hierarchy correctly identified and described; a specific, plausible, garage-relevant example given for each level.
5–7	Most levels correctly identified and described; examples given but one or two less specific to Con's Garage.
2–4	Some levels of the hierarchy identified; descriptions or examples generic or incomplete.
0–1	Minimal or no understanding of the ABC cost hierarchy.

QUESTION 2 TOTAL — 30 MARKS

QUESTION 3

Asia Ltd

Total: 30 marks

Asia Ltd, month of July: no opening stock, 6,000 units produced, 5,400 units sold, selling price €90. Unusually, both manufacturing overhead *and* advertising are split into variable (per-unit) and fixed (total) components. Candidates are required to (A) explain the terms 'variable costing' and 'absorption costing', (B) calculate unit product cost and closing stock under both methods, and (C) prepare full profit and loss accounts under both methods.

Part	Requirement	Marks
(A)	Explain 'variable costing' and 'absorption costing'; why profits may differ	6
(B)	Unit product cost and closing stock value under both methods	6
(C)	Profit and loss account for July – absorption costing and variable costing	18

Question 3(A) **6 marks**

Explain what is meant by the terms 'Variable costing' and 'Absorption costing'. In your answer, state why the operating profits resulting from the use of the two costing methods may differ.

Full Model Answer

Variable costing (marginal costing)

Variable costing values each unit of product at variable manufacturing cost only (direct materials, direct labour, and variable manufacturing overhead), treating fixed manufacturing overhead as a period cost, expensed in full in the period it is incurred regardless of how many units are sold.

Absorption costing

Absorption costing values each unit of product at full manufacturing cost, including a share of fixed manufacturing overhead absorbed into each unit produced. Fixed manufacturing overhead is therefore carried forward in inventory (as part of the cost of unsold units) until the unit is eventually sold.

Why the two methods' profits may differ

Because absorption costing defers fixed manufacturing overhead into closing stock while variable costing expenses it immediately, the two methods report different profit whenever production does not exactly equal sales in the period. When production exceeds sales (as at Asia Ltd this month), more fixed overhead is deferred into closing stock than is expensed, so absorption costing reports a higher profit than variable costing.

Mark Allocation

Marks	Criteria
5–6	Both terms accurately defined, correctly identifying the treatment of fixed manufacturing overhead as the key distinction; clear explanation of why profits differ, referencing the deferral of fixed overhead in stock.
3–4	Both terms defined with reasonable accuracy; explanation of the profit difference less developed.
1–2	One term defined well; the other only partially, or the reason for the profit difference not explained.
0	No meaningful explanation of either term.

Question 3(B)

6 marks

Asia Ltd, July: 6,000 units produced, 5,400 sold, no opening stock. Sales price €90/unit. Variable costs per unit: DM €21.00, DL €30.00, Manufacturing overhead €9.50, Advertising €5.50. Fixed costs: Manufacturing overhead €72,000, Advertising €16,500. Determine the unit product cost and the value of closing stock under absorption costing and under marginal costing.

Full Model Answer

Important distinction: Advertising (both its variable €5.50/unit and fixed €16,500 elements) is a **selling** cost, not a manufacturing/product cost, and is therefore never included in the unit product cost under either costing method — it is charged directly against profit as a period expense. Only direct materials, direct labour, and manufacturing overhead (variable and, under absorption costing, fixed) form part of the unit product cost.

Working – unit product cost

Item	€
Direct materials	21.00
Direct labour	30.00
Variable manufacturing overhead	9.50
Variable (marginal) costing unit product cost	**60.50**
Fixed manufacturing overhead per unit (€72,000 ÷ 6,000 units)	12.00
Absorption costing unit product cost (€60.50 + €12.00)	**72.50**

Working – closing stock value

Item	Value
Closing stock units (6,000 produced − 5,400 sold)	600 units
Closing stock value – variable/marginal costing (600 × €60.50)	**€36,300**
Closing stock value – absorption costing (600 × €72.50)	**€43,500**

Mark Allocation

Marks	Criteria
5–6	Advertising correctly excluded from unit product cost under both methods; both unit product costs and both closing stock values correctly calculated.
3–4	Substantially correct with a minor error, most commonly incorrectly including a component of advertising in the unit product cost.
1–2	Some correct workings but a fundamental misunderstanding of which costs belong in the unit product cost.
0	No meaningful correct workings.

Question 3(C) **18 marks**

Prepare a profit and loss account for July, using (i) Absorption costing; (ii) Variable costing.

Full Model Answer – (i) Absorption costing

Profit and loss account, July – absorption costing

Item	€
Sales revenue (5,400 units × €90)	486,000
Less: Cost of goods sold (5,400 units × €72.50)	(391,500)
Gross profit	**94,500**
Less: Variable advertising (5,400 units × €5.50)	(29,700)
Less: Fixed advertising	(16,500)
Operating profit (absorption costing)	**48,300**

Full Model Answer – (ii) Variable (marginal) costing

Profit and loss account, July – variable costing

Item	€
Sales revenue (5,400 units × €90)	486,000
Less: Variable cost of goods sold (5,400 units × €60.50)	(326,700)
Less: Variable advertising (5,400 units × €5.50)	(29,700)
Contribution margin	**129,600**
Less: Fixed manufacturing overhead	(72,000)
Less: Fixed advertising	(16,500)
Operating profit (variable costing)	**41,100**

Reconciliation: The €7,200 profit difference (€48,300 absorption − €41,100 variable) equals the fixed manufacturing overhead carried forward in the 600 units of closing stock under absorption costing (600 × €12.00 = €7,200), consistent with production (6,000 units) exceeding sales (5,400 units) in July and there being no opening stock to release.

Mark Allocation

Marks	Criteria
15–18	Both profit and loss accounts fully and correctly prepared, correctly separating variable and fixed advertising in both formats, with the correct contribution margin/gross profit subtotals and final operating profit figures.
11–14	Both accounts substantially correct with a minor error, e.g. in splitting variable/fixed advertising.
6–10	One account (absorption or variable) correct; the other contains significant errors, most commonly misclassifying advertising or fixed manufacturing overhead.
0–5	Limited understanding of either format; minimal correct workings.

QUESTION 3 TOTAL — 30 MARKS

QUESTION 4

Statement Evaluation

Total: 30 marks

Five independent statements covering prime vs conversion costs, breakeven/margin of safety/operating leverage verification, differential vs opportunity costs, fixed cost per unit behaviour, and the relevant range. For each statement, candidates must state whether they agree or disagree, with supporting calculations and explanation.

Part	Topic	Marks
(A)	Prime costs vs conversion costs	6
(B)	Bright Ltd – breakeven, margin of safety and operating leverage verification	6
(C)	Differential costs vs opportunity costs	6
(D)	Fixed cost per unit as activity level changes	6
(E)	The relevant range in decision-making	6

Question 4(A) **6 marks**

Prime cost consists of all direct manufacturing costs and is the same thing as conversion costs.

Full Model Answer

DISAGREE

The first part of the statement is broadly correct — prime cost is the total of direct manufacturing costs (direct materials, direct labour, and any direct expenses). However, prime cost is **not** the same thing as conversion cost.

> **Prime cost vs conversion cost**
> Prime cost = Direct materials + Direct labour (+ any direct expenses). Conversion cost = Direct labour + Manufacturing overhead. The two share direct labour in common, but differ in their other component: prime cost includes direct materials and excludes manufacturing overhead, while conversion cost includes manufacturing overhead and excludes direct materials. In Question 3, for example, Asia Ltd's prime cost per unit would be €21.00 + €30.00 = €51.00, while its conversion cost per unit would be €30.00 + €9.50 (variable) + €12.00 (fixed) = €51.50 — different figures, illustrating that the two measures are not interchangeable.

Mark Allocation

Marks	Criteria
5–6	Correct disagreement with accurate definitions of both prime cost and conversion cost, correctly identifying direct labour as the only common element and explaining why the two totals otherwise differ, ideally with a numerical illustration.
3–4	Correct disagreement with accurate definitions of both terms; the shared/differing components not explicitly explained.
1–2	Correct disagreement but with a vague or partially inaccurate definition of one term.
0	Incorrect verdict or no meaningful explanation.

Question 4(B) **6 marks**

Bright Ltd produces a single product. If total sales are 2,000 units, totalling €19,000; total variable expenses are €7,300 and total fixed costs are €9,360; then break-even point is 1,600 units, the margin of safety is 20% and operating leverage is 5 times.

Full Model Answer

AGREE

All three figures given in the statement are correct.

Working

Item	Value
Selling price per unit (€19,000 ÷ 2,000)	€9.50
Variable cost per unit (€7,300 ÷ 2,000)	€3.65
Contribution margin per unit	€5.85
Break-even point (€9,360 ÷ €5.85)	**1,600 units – correct as stated**
Margin of safety ((2,000 − 1,600) ÷ 2,000)	**20% – correct as stated**
Total contribution margin (2,000 × €5.85)	€11,700
Operating profit (€11,700 − €9,360)	€2,340
Operating leverage (€11,700 ÷ €2,340)	**5.0 – correct as stated**

Note: Unlike similar statements elsewhere in this series (where the break-even point and margin of safety check out but the operating leverage figure is deliberately wrong), all three figures in this statement are independently verified as correct here.

Mark Allocation

Marks	Criteria
5–6	All three figures independently recalculated and verified as correct; clear agreement verdict with full supporting workings.
3–4	Two of the three figures correctly verified; the third verified with a minor computational error.
1–2	Some correct workings but the verdict not clearly supported by all three figures.
0	Incorrect verdict (disagrees with the statement) or no meaningful workings.

Question 4(C)

6 marks

A difference in costs between any two cost alternatives is known as an opportunity cost.

Full Model Answer

DISAGREE

The description given in the statement is the definition of a **differential cost** (also called an incremental cost), not an opportunity cost.

> ### What a differential cost actually is
> A differential cost is the difference in total cost between two alternative courses of action being compared. It is a real, recorded (or recordable) cost difference and is directly relevant to choosing between the alternatives.

> ### What an opportunity cost actually is
> An opportunity cost is the potential benefit given up when one course of action is chosen over the next best alternative. Unlike a differential cost, it is never recorded in the accounting records at all — it represents a forgone benefit, not a difference in recorded costs.

Mark Allocation

Marks	Criteria
5–6	Correct disagreement with accurate definitions of both differential cost and opportunity cost, clearly identifying that the statement has the two terms swapped.
3–4	Correct disagreement with an accurate definition of one of the two terms.
1–2	Correct disagreement but with vague or partially inaccurate definitions.
0	Incorrect verdict (agrees with the statement) or no meaningful explanation.

Question 4(D)

6 marks

Fixed costs per unit decrease as the activity level rises and increase as the activity level falls.

Full Model Answer

AGREE

This is a correct description of fixed cost per unit behaviour within the relevant range.

> **Why this relationship holds**
>
> Total fixed costs remain constant within the relevant range regardless of the level of activity. Since fixed cost per unit is calculated as total fixed cost divided by the number of units of activity, spreading the same total fixed cost over more units (as activity rises) reduces the fixed cost per unit, while spreading it over fewer units (as activity falls) increases the fixed cost per unit. In Question 1, for example, Orla's €22,000 total fixed cost gives a fixed cost per unit of €8.80 at 2,500 units sold, but would be €11.00 per unit at only 2,000 units, or €6.29 per unit at 3,500 units — falling as volume rises, and rising as volume falls.

Mark Allocation

Marks	Criteria
5–6	Correct agreement with clear explanation of the inverse relationship between fixed cost per unit and activity level, with a numerical illustration.
3–4	Correct agreement with reasonable explanation; no numerical illustration given.
1–2	Correct agreement but with minimal or unclear explanation.
0	Incorrect verdict or no meaningful explanation.

Question 4(E) — 6 marks

The relevant range does not matter in organisational decision-making.

Full Model Answer

DISAGREE

The relevant range is central to sound organisational decision-making, not irrelevant to it.

Why the relevant range matters

The relevant range is the band of activity within which the assumed cost behaviour relationships (constant total fixed costs, constant variable cost per unit) are expected to hold. Every cost estimate, budget, or breakeven calculation — including all of the CVP analysis in Questions 1 and 2 of this paper — is only valid within some relevant range. Decisions that involve activity levels significantly outside the current relevant range (for example, a large increase in production requiring additional premises or equipment) risk being based on invalid cost assumptions if the relevant range is ignored: fixed costs may 'step up' to a new, higher level, and variable cost per unit may change (e.g. due to bulk discounts or capacity constraints).

Consequence of ignoring it

Ignoring the relevant range when making decisions about significant changes in scale — such as Orla's planned 30% sales increase in Question 1, or Con's Garage planning for a change in service volume in Question 2 — risks materially understating costs or overstating expected profit, since the linear cost relationships used in CVP analysis may simply not hold at a meaningfully different level of activity.

Mark Allocation

Marks	Criteria
5–6	Correct disagreement with a clear explanation of what the relevant range is and why ignoring it is risky for decisions involving significant changes in activity level, with reference to the consequences of getting this wrong.
3–4	Correct disagreement with reasonable explanation of the relevant range's importance.
1–2	Correct verdict stated but with minimal or unclear explanation.
0	Incorrect verdict (agrees with the statement) or no meaningful explanation.

QUESTION 4 TOTAL — 30 MARKS